

=====

QUESTION 1: AI-POWERED DRONE NAVIGATION

=====

Greedy Best-First Search
Path: Start -> B -> D -> Goal
Travel Cost: 11

A* Search
Path: Start -> B -> E -> Goal
Travel Cost: 9

Comparison
Greedy Search uses heuristic only.
A* uses actual cost + heuristic.
A* is preferred when safety, cost and optimality are important.

=====

QUESTION 2: SMART CITY TRAFFIC SIGNAL OPTIMIZATION

=====

Initial Signal Timings:
[30, 30, 70, 70]

Initial Traffic Cost:
415.0

Hill Climbing Result:
Signal Timings: [45, 50, 40, 55]
Traffic Cost: 0.0

Simulated Annealing Result:
Signal Timings: [45, 50, 40, 55]
Traffic Cost: 0.0

Genetic Algorithm Result:
Signal Timings: [43, 49, 42, 54]
Traffic Cost: 19.0

Recommendation:
Genetic Algorithm is suitable for large traffic optimization problems because it explores multiple solutions and reduces the risk of getting trapped in local optima.

=====

QUESTION 3: AUTONOMOUS MARS ROVER

=====

Initial Rover Knowledge:
{'Start': ['A', 'B']}

Rover is currently at: Start
Newly observed locations: ['A', 'B']

Rover is currently at: A
Newly observed locations: ['Start', 'C']

Rover is currently at: C
Newly observed locations: ['A', 'Goal']

Updated Internal Model:
Start -> ['A', 'B']
A -> ['Start', 'C']
C -> ['A', 'Goal']

Online Search Path:
Start -> A -> C -> Goal

Conclusion:
Online search is suitable because the rover does not have a complete map and must continuously update its knowledge while exploring.

=====

QUESTION 4: UNIVERSITY EXAM TIMETABLE USING CSP

=====

Valid Exam Timetable Found:
Mathematics -> Monday-Morning
Physics -> Monday-Afternoon
Chemistry -> Monday-Afternoon
ComputerScience -> Tuesday-Morning

CSP Constraints Satisfied:
1. No student has overlapping exams.
2. Each subject has one time slot.
3. Mathematics is scheduled before Computer Science.
4. Forward checking reduces invalid choices.

Recommended Strategy:
Backtracking Search with Forward Checking is appropriate for solving the timetable CSP.

QUESTION 5: STRATEGIC GAME AI

Minimax Result:

Best Game Value: 6

Alpha-Beta Pruning Result:

Best Game Value: 6

Evaluated Leaf Nodes: 9

Evaluation Function Example:

Game State Score: 380.0

Evaluation Factors:

1. Number of soldiers
2. Health of units
3. Resources collected
4. Distance to enemy
5. Territory controlled
6. Remaining time

Recommendation:

Minimax with Alpha-Beta Pruning is recommended because it produces the same optimal decision as Minimax while avoiding unnecessary branches.